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Assignment #4 | Mathematics - III (MTH 231) | Date: October 25, 2024
Branch - Electrical Engineering | Section - I and II

1. What is meant by an initial value problem? Explain the difference between an ordinary differential equation of order n and an initial value problem of order n.

2. Explain the "modifications" in the Modified Euler's method and how it helps to improve the accuracy of solutions as compared to the Euler's method.

3. Given the ordinary differential equation $y''' - yy'' + e^x y' - 4(xy)^{\frac{1}{6}} + 8 = 0$. Which of the following statements are True and which are False? Explain your answers.
   (a) There is no function y satisfying the given equation such that $y(0) = -\pi$.
   (b) There are exactly 3 functions y satisfying the given equation such that $y(0) = -\pi$.
   (c) There are infinitely many functions y satisfying the given equation such that $y(0) = -\pi$.

4. Using Taylor's series method, find $y(1.1)$ and $y(1.2)$ correct to four decimal places given that $y' = xy^{\frac{1}{3}}$ and y(1)=1. Further find the exact solution by solving the given ordinary differential equation and hence compute the error.

5. The air pressure in a chamber as a function of time is given by the equation $y' = x^2 + y^2$. If the air pressure at $t = 1$ is 2.3 bar, use Taylor's series to approximate the pressure at $t = 1.1$ and $t = 1.2$, correct to four decimal places.

6. Find the velocity at 0.1 and 0.2 given that $\dfrac{dv}{dt} = 3e^t + 2v$ and the body starts at rest, correct to four decimal places.

7. Calculate the value of y at 4.4 correct to four decimal places using Taylor's method for the initial value problem $5xy' + y^2 - 2 = 0$ with $y(4) = 1$.

8. Expanding using Taylor's theorem solve the given initial value problem correct to four decimal places for $x = 0.1$.
$$y' = 0.1(x^3 + y^2), \quad y(0) = 1$$

9. Use Picard's method of successive approximations to find the value of y at 0.1 correct to four decimal places given that $y' = \dfrac{y-x}{y+x}$ and $y(0) = 1$.

10. Solve correct to four decimal places using Picard's method of successive approximations for $x = 0.2$, given that $y' = x\sin(\pi y)$ and $y(0) = \dfrac{1}{2}$.

11. Given $y' = x^4 y + x$ and $y(0) = 3$, use Picard's method of successive approximations to find $y(0.15)$ correct to four decimal places.

12. Obtain y at 0.25, 0.5 and 1 correct to three decimal places where $y' = \dfrac{x^2}{y^2+1}$ given that $y(0) = 0$ using Picard's method of successive approximations

13. Solve the given initial value problem with $h = 0.2$ for $x = 0.6$ by Euler's method as well as Modified Euler's method.
$$y' = x + |\sqrt{y}| \quad , y(0) = 1$$

14. The differential equation governing the motion of a pressurised fluid along a narrow pipe of constant cross-sectional area is given by $v = log(x + v)$ with $v(0) = 2$ where v and x denote respectively the velocity and distance travelled by the fluid. Find the velocity correct to four decimal places for $x = 1.2$ and $x = 1.4$ using Modified Euler's method.

15. Using Modified Euler's method evaluate correct to four decimal places the value of y at 0.2 and 0.4, given that $y' = ye^x$ and $y(0) = 0$.

16. Using Modified Euler's method find $y(2.2)$ correct to four decimal places. given that $y' = -xy^2$ with $y(2) = 1$.

17. Using Runge-Katta methods of second and fourth order solve the given initial value problems correct to four decimal places.

    (a) $y' = 3x + \dfrac{y}{2}$ where $y(0) = 1$ at $x = 0.1$
    (b) $y' = \dfrac{y^2 - x^2}{y^2 + x^2}$ where $y(0) = 1$ at $x = 0.2$ and $x = 0.4$
    (c) $y' = \dfrac{x^2 + y^2}{10}$ where $y(0) = 1$ at $x = 0.1$
    (d) $y' = \dfrac{2xy + e^x}{x^2 + xe^x}$ where $y(1) = 0$ at $x = 1.2$ and $x = 1.4$
    (e) $y' = x + y^2$ where $y(0) = 1$ at $x = 0.2$
    (f) $y' = -2xy^2$ where $y(0) = 1$ using $h = 0.2$ at $x = 1$

18. Use Milne's predictor and corrector method to solve the initial value problem $y' = xe^{-x}(x^2 + y^2)$ with $y(0) = 1$ and find $y(0.4)$ correct to four decimal places.

19. Taking $h = 0.1$, solve the given initial value problem by Milne's method and find the value at $x = 0.5$ correct to four decimal places.
$$y' = 2e^x - y, \quad y(0) = 2$$

20. Solve the initial value problem $y'' - xy' - y = 0$ with initial conditions $y(0) = 1$ and $y'(0) = 0$ at the point 0.1, correct to four decimal places.